Abstract

Species Distribution Models (SDMs) are widely used to predict where species could potentially thrive based on environmental conditions. Their outputs, known as Habitat Suitability scores, are essential for applications like reintroduction planning, locating rare species, or assessing biodiversity responses to climate change. However, these scores often overestimate the actual areas occupied by species, as not all suitable sites are used due to biotic constraints, dispersal limitations, or historical contingencies. Over-prediction of habitat suitability reduces reliability for critical tasks like extinction risk assessments. To address this issue, spatial filtering methods are often applied to better approach realized species distributions. Most commonly, this involves clipping habitat suitability maps with binary spatial masks such as IUCN (International Union for Conservation of Nature) range maps or from minimal convex polygons around observations. Although easy to apply, these methods assumes full independence between the suitability map and the spatial mask, which can introduce inaccuracies and limit interpretability. Here, we introduce Permanence of Ratios, a geostatistical method that relaxes this assumption by adopting a conditional independence perspective. Using a Bayesian framework, Permanence of Ratios clarifies each map's role in the filtering process, avoids artificial renormalization, and produces results that are more accurate and interpretable. Using data from European bird species, we compare this method to traditional clipping across several spatial constraints, including expert range maps and observation-based techniques. Our results demonstrate Permanence of Ratios as a practical and effective tool to refine SDM outputs without additional data, making it beneficial for research in ecology and conservation.

GitHub repository link: https://github.com/anon-researcher-159487/Estar-copy

1 Introduction

The anthropogenic era begins with tremendous concerns and uncertainties regarding current and future human impacts on ecosystems and environments (Calvin et al. 2023). Although changes in species distributions and abundances have occurred throughout history,recent decades have witnessed unprecedented acceleration due to human activities. Species are increasingly experiencing rapid shifts in their range, primarily driven by climate change (Chen et al. 2011), significant habitat reductions due to human-induced landscape transformations

(Brooks et al. 2002) or, conversely, rapid expansions linked to invasions (Didham et al. 2007). These anthropogenic effects are critical due to their cascading impact on ecosystem functions and the global services that ecosystems provide, such as disease regulation, availability of water and food, and climate stabilization (Brondizio 2019). Consequently, wildlife conservation and restoration have emerged as global priorities, underscoring the necessity of accurate, spatially explicit mapping of species distributions to support informed conservation and restoration planning (Pollock et al. 2020).

Species Distribution Models (SDMs) are currently the tool of the choice to predict and map species distributions across space, particularly for locations where direct observations are lacking (Guisan and Thuiller 2005). Grounded on niche theory, these models have become popular over the past three decades due to their intuitive conceptual framework and relative accuracy. Typically, SDMs generate habitat suitability maps, assigning scores to indicate how favorable environmental conditions are for species occurrence. Higher scores suggest greater environmental suitability but do not necessarily equate directly to species presence. From these continuous maps, binary *potential range* maps are derived, explicitly indicating predicted presence or absence based on a suitability threshold. Both outputs have important applications, such as selecting optimal reintroduction sites (Jarvie and Svenning 2018), identifying invasion risks (Srivastava, Lafond, and Griess 2019), predicting dispersal pathways (Keeley, Beier, and Gagnon 2016), locating rare species (Williams et al. 2009), and forecasting climate change impacts on species ranges (Dormann 2007) (Chauvier-Mendes et al. 2024). Nevertheless, SDM outputs frequently overestimate actual species distributions (Velazco et al. 2020). This discrepancy arises because species typically occupy only subsets of suitable habitats, constrained by competition (Armstrong and McGehee 1980), absence of mutualist partners (Giannini et al. 2013), dispersal limitations (Chen et al. 2011), and historical contingencies (Zeisset and Beebee 2008). This over-prediction can undermine conservation planning, species protection assessments or extinction risk evaluations where accurately approximating realized species distributions is crucial (Bateman et al. 2013; Sinclair, White, and Newell 2010).

To mitigate this issue, methods for spatially constraining predicted habitat suitability or potential ranges have been developed, broadly categorized as *a priori* and *a posteriori* approaches.

A priori methods use spatially-explicit models, incorporating spatial variables alongside

environmental variables in the models (Allouche, Steinitz, et al. 2008; Domisch, Wilson, and Jetz 2016) or adopting mechanistic models that simulate colonization, extinction, and dispersal dynamics (Engler and Guisan 2009; Zurell et al. 2016).

In contrast, *a posteriori* methods apply spatial constraints as post-processing filters on existing SDM outputs. These typically involves three steps: (1) using SDM to generate a habitat suitability map using environmental predictors; (2) defining a spatial constraint or envelope based on expert-defined range maps (e.g. IUCN), or constructed using various methods, such as estimating accessible areas from environmental barriers (Cooper and Soberón 2018; Lessmann et al. 2016), incorporating expert-defined restrictions (Calixto-Pérez et al. 2018), utilizing resistance-based cost surfaces (Gherghel, Brischoux, and Papeş 2020), or employing geometric constraints like minimum convex hulls or spatial buffers (Kremen et al. 2008; Brown 2014). For a comprehensive review, see (Mendes et al. 2020) ; (3) incorporating habitat suitability maps with the spatial constraint to produce a constrained realized range map. The most common *a posteriori* method, "clipping" involves multiplying the SDM output by the spatial constraint, thereby setting values outside the spatial envelope to zero. However, this method implicitly assumes independence between the suitability and spatial constraint maps, an assumption often violated since both maps commonly derive from species occurrence data, leading to confounding effects. These issues often result in underestimated outputs, masked by artificial re-normalization to a [0,1] scale, thereby reducing interpretability.

To address these limitations, we introduce the **Permanence of Ratios** method (Journel 2002), originating from geostatistics, which relaxes the strict independence assumption with a more realistic conditional independence assumption. By using a Bayesian framework, this method explicitly incorporates the relationship between habitat suitability and spatial constraints, avoiding artificial renormalization and enhancing interpretability. Moreover, like all *a posteriori* methods, Permanence of Ratios leverages existing data, avoiding the need to construct entirely new models, thereby saving time and resources.

In this paper, we present the theoretical foundations of the Permanence of Ratios method and demonstrate its practical application in SDM contexts. Using European bird species as a case study, we illustrate how the Permanence of Ratios provide more interpretable and accurate outputs compared to traditional clipping approaches. Specifically, we evaluate its predictive performance using an independent dataset of breeding bird surveys in France,

comparing it against five spatial constraints, including four commonly used methods and a novel one, PseudoRange, introduced herein.

2 Framework

The Permanence of Ratios is a method for integrating diverse data sources using conditional probabilities to achieve more accurate spatial predictions of the focal variable, here the probability that a site is part of the species realized range.

Here, we outline the three critical steps involved in *a posteriori* spatial contraint methods and introduce the Permanence of Ratios method explicitly in step 3. First, we clarify the concept of habitat suitability and detail how to estimate it using Species Distribution Models (SDMs). Second, we describe various methods to create spatial envelopes used as spatial constraints. Finally, we explain how these spatial constraints are integrated with habitat suitability maps using the Permanence of Ratios method.

2.1 Habitat Suitability: methods and interpretation

Since the permanence of ratios relies on conditional probabilities, one need to translate commonly used outputs from Species Distribution Models (SDM) into a Bayesian probabilistic framework (Boxes 1 and 2 provide more information on Bayesian methods).

One of the key outputs from SDMs trained on environmental data is a prediction over a given spatial extent and resolution. This prediction, commonly named habitat suitability (HS) (Guisan, Thuiller, and Zimmermann 2017), is usually continuous and indicates locations (pixels, sites) where the environmental conditions align with those observed in the species' presence records. This score, reflecting the environmental similarity with the conditions used by the species, is typically normalized to a 0–1 range, with probability calibration varying based on the modeling algorithm (Elith* et al. 2006). For example, a logistic function is often used in Generalized Linear Models to map linear predictors to probabilities.

Despite their widespread use, the interpretation of the HS values remains ambiguous due to various factors such as species prevalence, algorithm type, and parameter setting (Albert and Thuiller 2008) . For clarity, we define habitat suitability explicitly as the likelihood of a site being suitable for a species based solely on environmental conditions, independent of species abundance or sampling biases. Thus, habitat suitability represents the conditional

probability $P(s \subset \Omega|Env)$, where Ω denotes the realized range—a collection of regularly occupied sites (e.g., feeding, sheltering, breeding) (Gaston and Fuller 2009). To ensure this interpretation, habitat suitability maps must exclude spatial information, species rarity aspects and sampling biases, solely reflecting environmental conditions

All species exhibit both true site prevalence (which is the proportion of occupied sites over the entire area), and a data prevalence (proportion of occupied sites within the training data). If data prevalence is balanced at 50%, HS values reflect purely environmental suitability, aligning with our definition of $P(s \subset \Omega|Env)$. The model is not considering potential rarity of the species, absences and presences are evenly represented in the training process.

However, if data prevalence is not 50%, the overall output values will be positively dependent of the data prevalence. For instance, imagine a species that displays low abundances and is therefore rarely observed, and so even at presence sites. If the data prevalence reflects this bias, the regression model will reduce overall prediction values to reduce false positive errors (because absences are more represented that presences), leading sometimes to low predicted values even at known presence sites that corresponds to suitable habitats. This is not anymore what we expect by habitat suitability. Indeed, this output is more linked with the probability of observation of the species: even at sites inside the realized range, if the species is rare, we have lower probability to observe it.

To use our method, and also to maintain a good interpretability, we recommend balancing prevalence at 50% in the training data to stick with our definition of habitat suitability. In the Bayesian framework, a probability value of 0.5 is interpreted as the maximum incertitude concerning the outcome of the event. Based on that, the best input would be the voting result from ensemble modeling that aggregate multiple predictions, thus offering robust estimates of habitat suitability (Araújo and New 2007) (Thuiller et al. 2009). In that case the 0.5 value is literally the maximum uncertainty, with half of the models agreeing to say that the site is in the realized range, and the other half disagreeing. This would be theoretically a closer estimation of $P(s \subset \Omega|Env)$ than the one obtain with habitat suitability.

To summarize, habitat suitability is the output of an SDM only taking into account environmental variable, independent as much as possible from information on location and observations processes. In that case, this habitat suitability score could be a good estimator for $P(s \subset \Omega|Env)$, which is the quantity truly wanted. In the case where several SDMs

have been created (for example varying the algorithm used), the result of the voting process involving these models is potentially an even better estimator of $P(s \subset \Omega|Env)$.

As such habitat suitability maps are an approximation of $P(s \subset \Omega|Env)$. Similarly, we further assume that spatial envelopes, such as those derived from observation data, approximate the conditional probability that a site belongs to the realized range, given its geographic coordinates only.

$$Habitat\ Suitability\ value\ at\ site\ s \simeq P(s \subset \Omega|Env) \quad (*)$$

$$Spatial\ Envelope\ value\ at\ site\ s \simeq P(s \subset \Omega|x, y) \quad (*^2)$$

Here, (x, y) represent geographic coordinates, $|$ indicates conditional information and $P(\cdot)$ denotes the probability of an event. For instance, $P(A|B)$ means "the probability that A is true if we know B". (*) and ($*^2$) are our fundamental working assumptions, they form the basis of the proposed method and are further discussed in the Discussion section.

2.2 Creating a spatial envelope and estimation of $P(s \subset \Omega|x, y)$.

2.2.1 Different methods to build a spatial envelope

A variety of post-processing can constrain habitat suitability maps from SDMs. We briefly describe four commonly used methods to produce a spatial envelope for a given species that will later be used as a spatial filter and a new one that we propose here. A spatial envelope is an ensemble of selected regions in a form of a binary map (1 for selected sites, 0 for non-selected). We will compare all the different spatial envelopes described below in the case study:

1. **Range maps**: When available (e.g., vertebrates), the extent of occurrence or range maps, such as those from the IUCN Red List database, are often used as a binary mask to restrict habitat suitability predictions.
2. **Occurrences-Based Restriction (OBR)**: Habitat Suitability predictions are binarised, here using the Otsu thresholding method (Otsu 1979). The envelope is defined as regions with occurrences (S_1) and nearby suitable regions without occurrences (S_0) that lie within a threshold distance T. The threshold T is chosen as the maximum of

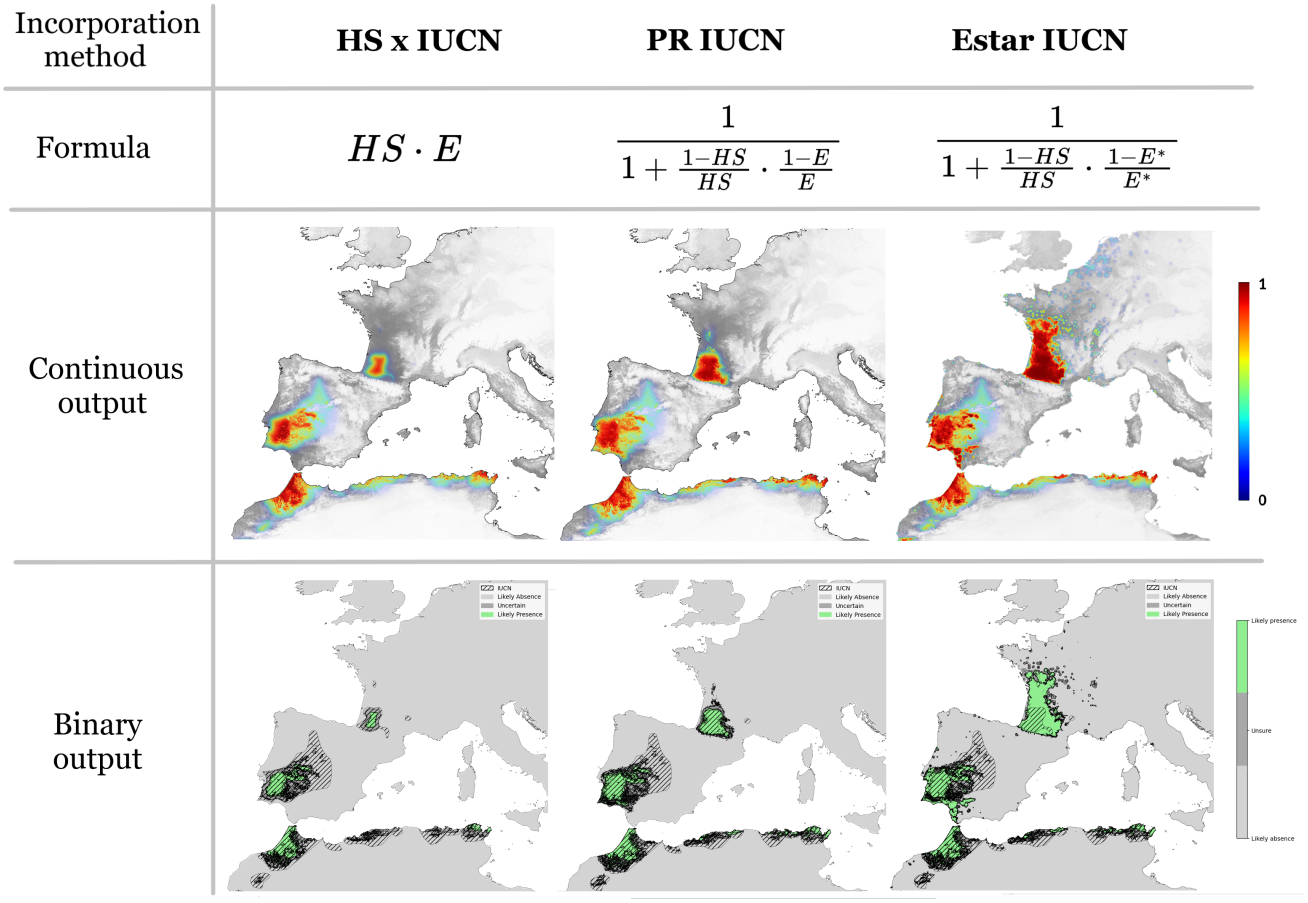


Figure 1: **Comparison between the three incorporation methods**. E is the normalized smoothed spatial envelope obtained from the binary envelope computed either by using MCP, OBR, LR, IUCN or PseudoRange method. Here the IUCN range extent is used for the maps shown. The full framework to compute E^* in the ESTAR method, is displayed in Fig. 2. The different maps depict predictions using the different incorporation methods for the blacked winged kite (*Elanus caeruleus*).

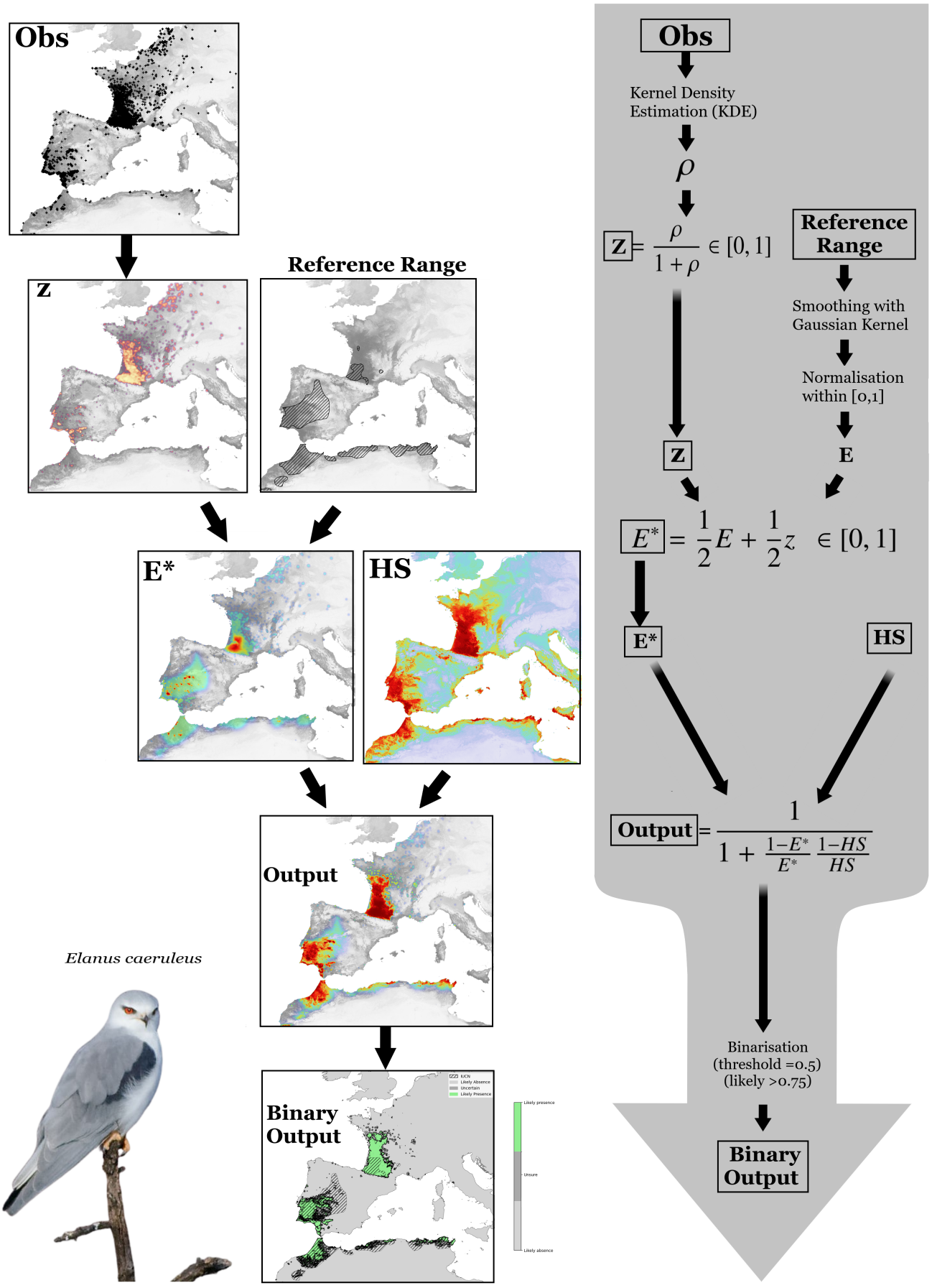


Figure 2: **Framework ESTAR:** Three inputs are provided: the habitat suitability, which is considered here as an estimate of $P(s \subset \Omega|Env)$, a spatial envelope as a reference range, here the IUCN range extent is shown, and the occurrences of the species. First, using the method described before, the occurrences points are used to build a continuous occurrence information, with kernel density estimation and a well-chosen transformation. Then, the reference range (spatial envelope) after being smoothed and normalized (E), is mixed with the occurrence information z with a linear combination to get E^* : an estimation of $P(s \subset \Omega|x, y)$. Finally, our estimations of $P(s \subset \Omega|Env)$ and $P(s \subset \Omega|x, y)$ are combined using permanence of ratios to get the final output: $P(s \subset \Omega|Env \cap x, y)$.

the minimal distances between pairs of occurrence points (typically around 300 km for European bird species).

3. **LR (Lower quartile Restriction)**: Similar to OBR, but only the closest 25% of suitable but unoccupied patches (S_0) from S_1 patches are included in the final envelop.

4. **MCP (Minimum Convex Polygon)**: The most commonly used *a posteriori* method to delineate a spatial envelop. Habitat Suitability predictions are binarised, here using Otsu thresholding (Otsu 1979). For each continuous suitable region containing at least three observations, a minimum convex hull is computed. The spatial envelop is defined as the union of all such polygons.

5. **PseudoRange**: A new method proposed here that combines information on species observations and the (eco)regions in which they occur to build the envelop, while accounting for sampling bias (this method does not use the habitat suitability to build the envelop). First, the study area is divided into ecologically meaningful subregions—e.g., ecoregions that represent homogeneous bioclimatic areas. To build the spatial envelope, subregions are selected and weighted based on two factors: (1) The observation anomaly, which the ratio between the number of focal-species occurrences in a given subregion and the number of occurrences we would expect if the species were distributed proportionally to a reference taxon. In our case study, the aggregation of all bird occurrences serves as the reference (i.e., total sampling effort). This ratio accounts for sampling-effort differences across subregions. (2) The coverage, which measures the spatial uniformity of focal-species occurrences within each subregion. We divide the total area of the subregion by the number of focal-species occurrences, then draw a disk of that area around each occurrence point. The union of these disks gives a total "covered" area, and the coverage ratio is defined as the total area covered by disks divided by the total subregion area. Clustered occurrences produce overlapping disks (reducing the coverage ratio), whereas well-distributed occurrences produce fewer overlaps (increasing the coverage ratio). Each subregion is assigned a score equal to the product of its observation anomaly and its coverage. Subregions that are only partially occupied by the focal species will have low coverage (and thus a lower score or weight). This selection procedure addresses sampling bias and mitigates the risk of overprediction when including large subregions with limited actual occupancy. The final spatial envelope comprises the selected subregions, each weighted by the product of its coverage

and observation anomaly.

Although $P(s \subset \Omega|x, y)$ is a probability and therefore take values between 0 and 1, the above-mentioned envelop methods, except PseudoRange, permit only to select a binary extent, which is a rather *sharp* estimation of $P(s \subset \Omega|x, y)$. An alternative, which is also more adapted to Permanence of Ratios, would be to consider smooth - continuous - boundaries.

2.2.2 Estimation of $P(s \subset \Omega|x, y)$ from the spatial envelope

To approximate a continuous probability, we propose to smooth the binary spatial envelope with a Gaussian kernel, normalizing the resulting continuous map between 0 and 1. We refer to this normalized, smoothed, envelope as: E (for Envelope). See "smoothing" in Fig. 2.

At this stage, E can directly approximate $P(s \subset \Omega|x, y)$ and be incorporated with habitat suitability, HS. Two incorporation methods are considered here: either the basic clipping ("HS x E") or the Permanence of Ratios ("PR E"). For instance, using simple product combination between HS and an IUCN-derived E produces the models HSxIUCN (product), using instead Permanence of Ratios yields to PR IUCN . Combining two incorporation methods and five envelope-construction methods results in ten spatial constrain variations.

Furthermore, we propose refining E with small-scale details from occurrence density, producing a modified envelope termed E^*. The Permanence of Ratios approach applied to E^* defines the model family "ESTAR". This yields five additional spatial constrain variations ("ESTAR IUCN", "ESTAR MCP", "ESTAR LR", "ESTAR OBR", and "ESTAR PseudoRange"), for a total of 15 different variations evaluated in our case study.

2.2.3 Refined estimation of spatial envelop: ESTAR method

The ESTAR approach refines spatial envelopes by incorporating fine-scale occurrence densities via spatially adaptive Kernel Density Estimation (KDE, see "Kernel Density Estimation" in Fig. 2). To address variations in sampling intensity, the width of the kernel is adapted locally: sparse sampling requires larger kernels to prevent spotty effects, while densely sampled regions require narrower kernels to maintain detail. To deal with this issue, a distance-based network, derived from occurrences and thresholded by the median pairwise distance between occurrences pairs, is partitioned into regions of homogeneous sampling density using commu-

nity detection (Louvain's algorithm). Within each community, the kernel standard deviation is set as the median of minimal pairwise distances.

The resulting KDE yields a continuous density map $\rho(s) > 0$, subsequently transformed to reflect presence likelihood. Few close enough observations is a good sign of species presence, but confidence quickly saturates with increasing occurrence numbers (confidence does not increase much when observing 1000 occurrences instead of 998 in a region, but greatly increases if we observe 3 points instead of one). We thus apply a bounded, nonlinear transformation: (see Fig. 2):

$$z = \frac{\rho}{1+\rho} \in [0,1]$$

We combine the smoothed envelope (E) and density-derived map (z) into the refined envelope E^* as a linear combination:

$$P(s \subset \Omega | x, y) \simeq E^* = \frac{1}{2}E + \frac{1}{2}z \quad \in [0,1]$$

(see Fig.2)

This combination ensures moderate confidence (0.5) when either spatial envelope or occurrence density alone supports occupancy, whereas coinciding high-confidence evidence (both dense observations and envelope inclusion) yields stronger occupancy likelihood. When a site is within the envelop but without known observations, the likelihood that the species actually uses that site is uncertain (flat prior). The same applies for an area of dense observed occurrences outside the envelope. The confidence is moderate since those occurrences could be true occurrences or be due to museum collections, wrong identifications, or vagrant or migratory individuals. Instead, when both envelop and density maps agree (dense cluster of occurrences within the envelope), there is high chance that the species really uses the site and the overall value can rise above 0.5 passing from a flat prior to an informative prior at that site. We interpret E^* as a robust approximation for $P(s \subset \Omega | x, y)$.

2.3 Incorporation phase: Permanence of Ratios vs. Clipping

The Permanence of Ratios method transforms probabilities, and their associated numerical instabilities (Box 1 and 2), into numerically stable ratios:

$$d_A = \frac{1 - P(A)}{P(A)}$$

Ratio can be considered as a transformation from probabilities between 0 and 1, to a distance from full confidence regarding the occurrence of A. If $P(A) \to 0$ this is very unlikely that A occurs, so $d_A \to +\infty$, and if $P(A) \to 1$ this is nearly certain that A occurs so $d_A \to 0$.

The interesting property of this transformation is that ratios are known to be numerically more stable. The Permanence of Ratios assumption stipulates that the relative decrease in uncertainty upon incorporating new information (C) remains constant irrespective of prior knowledge (B). In other words, the order in which an additional information is added does not matter, the level of information won is the same at the end. This can be written mathematically as:

$$\frac{d_{A|C\cap B}}{d_{A|B}} \simeq \frac{d_{A|C}}{d_A} \quad (*^3)$$

or equivalently:

$$\frac{d_{A|B\cap C}}{d_{A|C}} \simeq \frac{d_{A|B}}{d_A} \quad (*^3)$$

Assuming this assumption, the expression of $P(A|C \cap B)$ is straightforward:

$$P(A|B \cap C) = \frac{d_A}{d_A + d_{A|B} \cdot d_{A|C}}$$

Explicitly:

$$P(A|B \cap C) = \frac{\frac{1-P(A)}{P(A)}}{\frac{1-P(A)}{P(A)} + \frac{1-P(A|B)}{P(A|B)} \cdot \frac{1-P(A|C)}{P(A|C)}} \quad (4)$$

(4) uses $P(A|C)$, $P(A|B)$ and $P(A)$ and has the particularity to enforce a result in [0,1], which makes it robust to uncertainties on $P(A|C)$, $P(A|B)$ or $P(A)$. In practice, in the following framework, we define species' potential range as an estimation of $P(s \subset \Omega|Env)$, and a spatial constrain as an estimation of $P(s \subset \Omega|x, y)$. In our framework, we will use (4) with A $= s \subset \Omega$, B$=Env$ and C$=x, y$.

Keeping the same notations, what happens when clipping the potential range (i.e. habitat suitability in our case) by the spatial extent? A clip is a multiplication between both maps often followed by a renormalization giving:

$$Clip = \frac{P(s \subset \Omega|Env) \cdot P(s \subset \Omega|x,y)}{max(\{P(s \subset \Omega|Env) \cdot P(s \subset \Omega|x,y), s \in S\})}$$

But $P(s \subset \Omega|x,y)$ cannot being multiplied directly with $P(s \subset \Omega|Env)$ to give $P(s \subset \Omega|Env \cap x,y)$ without assuming that both maps are independent, which is unrealistic. On that case and only on that case, using (3):

$$P(s \subset \Omega|Env \cap x,y) = \frac{P(s \subset \Omega \cap Env \cap x,y)}{P(Env \cap x,y)}$$
$$= \frac{P(s \subset \Omega|Env) \cdot P(s \subset \Omega|x,y)}{P(s \subset \Omega)}$$

This underlies that clipping in this interpretation is close to the robust formula of incorporation, if both maps are independent of each other, but do not correspond entirely to the latter, due to the denominator. We propose to use (4) instead to deal with most of these problems, but converting them to the hypothesis $*^3$.

Assumption $*^3$ leads to a stable formula (4), which preserves the interpretation of the output map. Moreover, $*^3$ is more reasonable than the full independence hypothesis, despite being difficult to prove in practice (IUCN range maps surely rely on observed occurrences, but how much is difficult to estimate), (4) is often used in geostatistics with good results, see (Isunza Manrique et al. 2023) for a recent example. Other methods of multiple source integration exists, such as the tau-method or the log ratio method (Hong, M. Ortiz, and V. Deutsch 2007), but contrarily to Permanence of Ratios, they rely on further parameterizations and hypotheses.

In the following, $P(s \subset \Omega)$ is assumed to be a constant over the entire map. $P(s \subset \Omega)$ is fixed hereafter at 0.5, to allow simple interpretation when building $P(s \subset \Omega|x,y)$, see (Deutsch 2020) to see the effect of $P(A)$ on the output giving $P(A|B)$ and $P(A|C)$. If $P(s \subset \Omega)$ is 0.5, it gives a simple interpretation to $P(s \subset \Omega|x,y)$ and $P(s \subset \Omega|Env)$ values: if $P(A|.)$ has a value of 0.5, we have no particular confidence to say if the site is part of the realized range or not, regarding at the information ".". Higher (resp. lower) values means more confidence to attest the site is part of the realized range (resp. not a part of it).

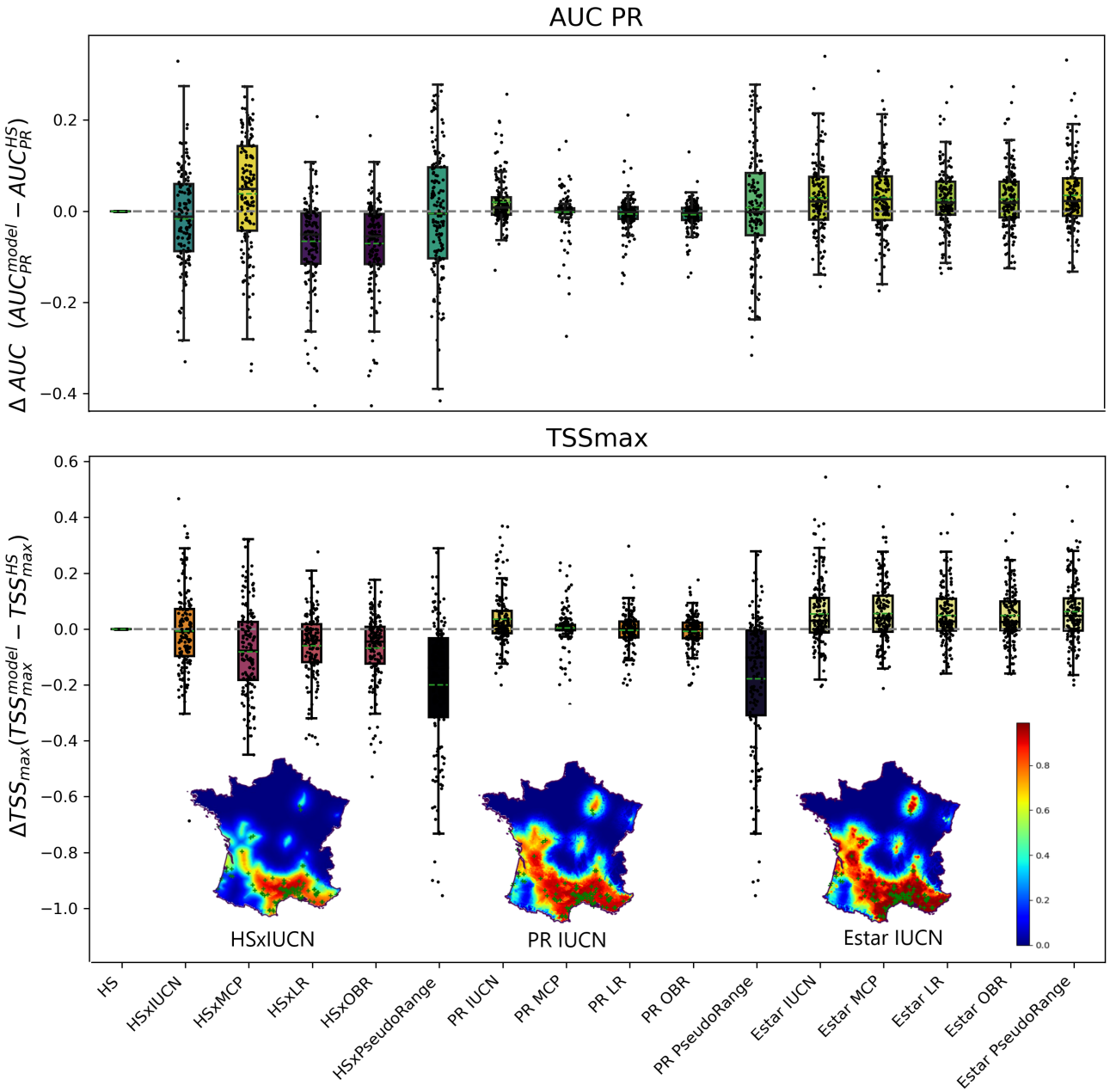


Figure 3: Change in TSS_{max} and AUC scores for all combinations of spatial constraints and incorporation methods compared to corresponding raw habitat suitability. TSS max is the maximum True Skill Statistics attained by the model for an optimized threshold. AUC PR is the Area Under the Curve under the Precision-Recall curve. Grey dotted line represents a score equal to the unconstrained habitat suitability. Green dotted lines correspond to means. An example of map prediction is shown for different spatial incorporation methods. Green crosses show presence validation points from STOC data surveys (Jiguet et al. 2012), (absences are not shown here). Results are based on species distribution models of 149 bird species, present in France when breeding, and with more than 20 occurrences. All models have been computed at European scale, France constituting a close up for evaluating local performance of the continental model.

3 Case Study: European species distributions

3.1 Difference of prediction accuracy when using Permanence of Ratios compared to clipping: application to European birds species.

European vertebrates, including birds, mammals, amphibians, and reptiles, are extensively studied, enabling the creation of reliable reference ranges through standardized bird surveys (Rands 2000) and assessments by IUCN (Butchart et al. 2004), (Stuart et al. 2004), (Schipper et al. 2008), (Böhm et al. 2013).

We compared the five outlined spatial envelope methods (IUCN, MCP, LR, OBR, PseudoRange) combined with the three detailed incorporation approaches: standard clipping, Permanence of Ratios, and ESTAR. Habitat suitability (HS) maps used in this study were generated uniformly across species, following a standardized modeling pipeline.

Specifically, we modeled European breeding bird species distributions (149 species) using presence-only occurrence data from GBIF projected onto a 1 km^2 grid over Europe. We generated two complementary sets of pseudo-absence data: (i) pseudo-absences sampled within each species' known range, and (ii) background points sampled outside their range. For each species, five datasets were created and further partitioned into five spatial cross-validation folds. Species distribution models (SDMs) were developed using an ensemble of three algorithms (Random Forest, XGBoost, Neural Networks) and incorporating key environmental predictors (climate, soil, terrain, hydrography, and land use). This yielded an ensemble of 75 models per species, from which only models achieving a TSS > 0.4 were retained. Final habitat suitability maps were obtained by averaging predictions across these selected models.

For all bird species analyzed, we applied consistent parameterization: each selected spatial extent (E) was smoothed using a Gaussian kernel with a standard deviation (σ) of 50 km, which is roughly the precision of such envelope at this study scale. For ESTAR, occurrences were filtered to include only observations within the IUCN breeding range to minimize biases introduced by migratory behavior and observation noise.

Spatial constraints were subsequently applied using either the traditional clipping or the Permanence of Ratios method. In the clipping approach, habitat suitability maps were

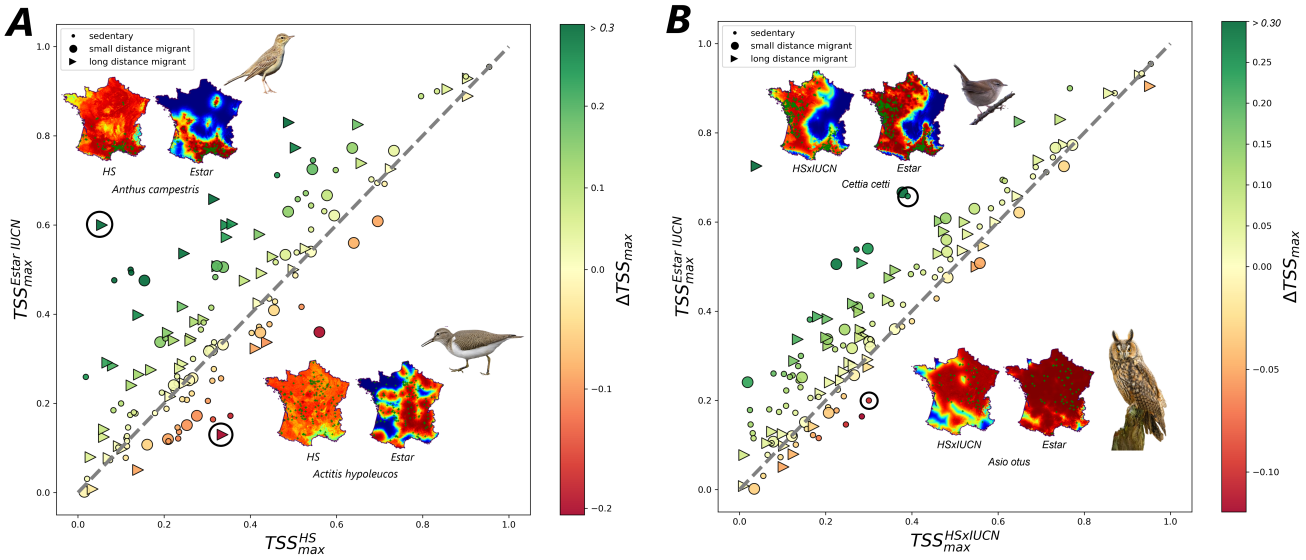


Figure 4: **(A) Close-up comparison of TSS scores between the ESTAR IUCN model and raw habitat suitability (HS)**. Species migratory classifications (long- and short-distance migrants, sedentary) are based on [sorkova$_h$ora$k_2$018].**(B) Comparison of TSS scores between the ESTAR IUCN model and the clipping meth**

multiplied directly by the reference smoothed envelope and rescaled linearly between 0 and 1. Conversely, the Permanence of Ratios method employed Equation (4) described previously. Model performances were assessed using two metrics: the maximum True Skill Statistic (TSS_{max}) and the Area Under the Precision-Recall Curve (AUC_{PR}) (Fig. 3).

It is important to underline that all spatial constraints considerably reduce over-prediction at the European scale compared to raw habitat suitability. For instance, as illustrated by Fig. 2, predictions originally overestimating suitability in areas such as the Italian peninsula, Balearic Islands, and Sicily were effectively reduced to near-zero suitability after constraint ($P(s \subset \Omega | Env \cap x, y)$), aligning well with actual recorded distributions. This better differentiation between presence sites and absences leads to a more accurate final binary map.

Our comparison of model performances specifically focuses on local accuracy, exemplified by the regional scale in France. For this, we evaluated predictions leveraging the comprehensive spatial coverage provided by STOC data in France, which is fully independent from the GBIF occurrence data used to build the SDMs. Since model evaluation metrics (AUC_{PR} and TSS_{max}) are sensitive to the prevalence of validation datasets (Allouche, Tsoar, and Kadmon 2006; Wunderlich et al. 2019) and to ensure comparability across species, we enforced a balanced prevalence of 50% for all validation sets through random resampling of validation locations (Fig. 3).

3.1.1 Comparison of local model performances

Compared to the original, unconstrained habitat suitability maps, the clipping method resulted in an average decline in local prediction accuracy. Conversely, the Permanence of Ratios method (using envelope E alone) maintained prediction accuracy comparable to the original habitat suitability maps. Importantly, the ESTAR approach (incorporating the refined spatial envelope E^* with Permanence of Ratios) significantly enhanced predictive performance relative to clipping, both in terms of AUC_{PR} (mean improvement: $\Delta AUC_{PR} = +0.05$, $p \ll 0.0001$) and TSS_{max} (mean improvement: $\Delta TSS_{max} = +0.13$, $p \ll 0.0001$), with statistical significance assessed using a Wilcoxon signed-rank test.

Focusing on the best-performing ESTAR variant (ESTAR-IUCN), 65% of species exhibited improved local predictions relative to unconstrained habitat suitability (Fig. 4A). The most pronounced gains were observed among long-distance migratory species, likely due to the inherently noisy spatial occurrence patterns associated with migratory behavior, resulting in less informative habitat suitability maps. The spatial constraint thus substantially enhanced predictions in these cases. Moreover, ESTAR IUCN outperformed the simpler HS $\times$ IUCN approach for approximately 75% of species (Fig. 4B).

We further explored the relationship between model improvements and species-specific sampling effort (number of occurrences). Contrary to expectations, the observed improvement in ESTAR IUCN relative to HS $\times$ IUCN (ΔTSS_{max}) was not significantly correlated with sampling effort ($p = 0.15$). However, higher sampling effort negatively impacted overall predictive accuracy (TSS_{max}; $p < 0.0001$), suggesting increased sampling often introduces dispersal noise rather than improving model accuracy. This indicates that ESTAR's improvements are species-specific rather than purely driven by sampling intensity, benefiting both abundant and rare species.

3.1.2 Case of a fast-expanding species

We additionally tested ESTAR on the black-winged kite (*Elanus caeruleus*), a rapidly expanding non-migratory species in France. First recorded breeding in 1990, the species has since spread extensively across southwest France, and its latest IUCN breeding range (2018) is already considerably outdated and underestimated. As demonstrated in Fig. 2, ESTAR successfully captured recent distributional expansions despite an outdated reference range,

owing to the integration of accumulated occurrence evidence. In contrast, simple clipping or Permanence of Ratios approaches could not account for such rapid distribution changes. The transformation applied to occurrence density (z) effectively down-weighted isolated occurrences, enabling realistic predictions of currently occupied areas despite noisy observational data. This result shows a potentially interesting application to fast changing distributions, even when occurrences data are highly noisy.

4 Discussion

Over-prediction remains a pervasive challenge in species distribution modeling. Although numerous strategies exist to mitigate this issue, they remain underutilized in practice (Velazco et al. 2020). Over-predictions in SDMs can stem from incomplete environmental niche modelling (Guisan, Chevalier, et al. 2025) or an inadequate representation of spatial limitations experienced by the species such as dispersal capacities (Chen et al. 2011) and historic range heritage (Zeisset and Beebee 2008)). In this work, we specifically addressed spatial limitations. Common approaches to reducing spatially-induced over-prediction consist of applying specific strategies to generate pseudo-absences (Senay, Worner, and Ikeda 2013), modeling unknown spatial dependencies explicitly (Redding et al. 2017), or directly incorporating spatial variables like Gaussian distances (Allouche, Steinitz, et al. 2008) or spatial eigenvectors (Diniz-Filho and Bini 2005) as predictors. These methods are considered *a priori*, integrating spatial information directly into the SDM. However, *a priori* methods have a critical limitation: the resulting models are often not easily transferable for future projections due to the spatial specificity of their training data (Rousseau and Betts 2022). Additionally, integrating spatial information within the model itself may entangle distinct types of information, complicating interpretation. By contrast, *a posteriori* approaches clearly separate environmental suitability and spatial constraints, aligning naturally with community assemblage theory (Cody and Diamond 1975) and providing better control over model biases.

Among *a posteriori* methods, a common and simple strategy is clipping SDM outputs using reference ranges. However, we demonstrated that this method often compromises local predictive accuracy and interpretability.

To overcome this issue, we introduced the Permanence of Ratios (Journel 2002), a geostatistical approach characterized by its conceptual simplicity and computational efficiency

(Equation 4). We showed that Permanence of Ratios offers clearer assumptions, improved interpretability, and enhanced local predictive accuracy compared to traditional clipping, without additional computational costs.Furthermore, we proposed ESTAR, an advanced refinement of Permanence of Ratios, integrating fine-scale occurrence information. Our results indicate that ESTAR further enhances local predictive accuracy, often surpassing raw habitat suitability predictions. ESTAR uniquely allows predicted range to extend beyond the reference boundaries when supported by occurrence data, making it particularly valuable for rapidly expanding species. Although ESTAR is computationally more demanding, it effectively handles for non-equilibrium distributions, mitigates sampling biases, and enhances small-scale distribution details.

4.1 Improvements and Limitations

The Permanence of Ratios approach, due to its computational simplicity and improved accuracy, is suitable for routine use when constraining habitat habitat suitability predictions. Ecological research often involves constrained resources, leading to preference for simpler, less accurate methods. Acknowledging this, we advocate adopting Equation (4) for improved constraints without additional computational cost.

Despite its computational complexity, ESTAR significantly enhances local details of species distributions and was shown here to be especially effective for species with non-equilibrium distributions. Although ESTAR uses occurrences to refine spatial constraints, it remains robust to common issues such as misidentification and sampling biases. This robustness arises from transforming occurrences density into a latent variable z, which displays a remarkably stable output despite heterogeneity in sampling effort.

However, ESTAR may still be affected by highly clustered erroneous occurrences (e.g., museum collections), potentially causing localized inaccuracies. Careful preprocessing of occurrence data should mitigate these risks.

Our findings on well-studied taxa, such as birds, are particularly noteworthy. Even when precise reference ranges (e.g., IUCN ranges) exist, applying Permanence of Ratios significantly improves model accuracy. This method thus holds promise for taxa with less accurate reference ranges, such as invertebrates, where gains in accuracy might be even more substantial.

In this study, occurrences were typically filtered by BirdLife breeding ranges due to birds' migratory behaviors. This potentially limited the method's utility, as demonstrated by our results for the rapidly expanding black-winged kite. Future research should explore ESTAR's potential for sedentary species, such as amphibians and reptiles, where it could yield even greater improvements.

4.2 Validity of Assumptions

Habitat suitability can have different interpretations depending on the algorithm used and the explanatory variables selected. The resulting scores can reflect probabilities of presence, observation probabilities, or conditional probabilities of presence given environmental conditions. Such ambiguity complicates SDM interpretation, especially when implicit information such as spatial or abundance biases is present..

To satisfy assumptions $(*)$ and $(*^2)$, we recommend a two-step modeling approach: First, include only environmental variables into SDMs, while minimizing the effects of species rarity (i.e. the inherent probability of observing the species in the region) and spatial dependencies. Under these conditions, the SDM output can be interpreted as an estimate of the conditional probability of a site for being part of the realized range: $P(s \subset \Omega|Env)$. In a second step, incorporate only spatial information—i.e., which regions of the study area are known to be inhabited by the species, independently of their environmental characteristics. This produces confidence maps estimating: $P(s \subset \Omega|x, y)$. $(*^2)$.

This separation contrasts with the common approach of modeling all variables simultaneously. By estimating $P(s \subset \Omega|Env)$ and $P(s \subset \Omega|x, y)$ separately, we can better identify biases and clarify the meaning of each output map, while still being able to project the environmental part in future conditions if needed.

The two resulting maps are then combined using the *Permanence of Ratios* method, which relies on a key assumption: the gain of information about the species' realized range is the same regardless of the order in which environmental and spatial information is added. Though challenging to empirically validate, this assumption has practical utility in geostatistics and represents a less restrictive alternative than the full independence assumption required by clipping methods.

4.3 Recommendations

We strongly recommend the adoption of the Permanence of Ratios formula (Equation 4) over the conventional clipping approach. This transition requires no additional computational resources yet significantly improves local predictive accuracy and interpretability.

When computational resources allow, we further encourage the adoption of ESTAR. The modest increase in computation time is justified by enhanced flexibility and predictive accuracy, particularly when dealing with incomplete reference ranges or rapidly shifting distributions.

Lastly, interpretability of habitat suitability scores is critical. Ideally, habitat suitability scores should remain as independent as possible from observational biases and species rarity-commonness. While many biases are deeply embedded in occurrence data, particularly those derived from citizen science databases, one major avoidable bias is the failure to correct for sampling prevalence during SDM training. If the training dataset is not balanced in terms of presence and absence (or pseudo-absence) records, habitat suitability values will be linked with species rarity-commonness, introducing systematic bias into the constraint process.

To mitigate this issue, we advocate for maintaining a balanced 0.5 prevalence ratio in SDM training data whenever possible. Additionally, we suggest using ensemble models rather than single-SDM outputs, as ensemble approaches are theoretically more robust approximations of $P(s \subset \Omega|Env)$ than individual model outputs (Dormann et al. 2018).

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